

Supplementary GC-PROVE Appraisal Checklist

Appraisal aid only; not a formal risk-of-bias tool or reporting guideline.

Problem anchoring 0: Pipeline- or gene-set-driven study without a clear question 1: Broad biological theme but vague clinical or mechanistic target 2: Explicit clinical, biological, or validation question with intended claim type Reviewer question: What decision, mechanism, or validation gap does the analysis address?

Resource provenance 0: Dataset names missing or unreconstructable 1: Accessions reported but filtering, endpoints, or overlap incompletely described 2: Accessions, versions, sample filters, endpoint definitions, missingness, and cohort overlap are auditable Reviewer question: Can another analyst reconstruct the cohort?

Omics workflow transparency 0: Preprocessing, feature selection, or model formula unavailable 1: Partial workflow reported, but code, versions, or tuning details incomplete 2: Normalization, gene mapping, batch correction, feature selection, model formula, coefficients, software versions, and code are provided Reviewer question: Are preprocessing and modeling choices reproducible?

Validation hierarchy 0: Discovery cohort only or validation after model rebuilding 1: External or biological support present but model/thresholds are not fully locked 2: Locked external validation plus relevant tissue, protein, spatial, functional, or clinical support proportional to the claim Reviewer question: Is the claim appropriately limited by its validation level?

Evidence restraint 0: Conclusions imply clinical readiness from retrospective public data 1: Limitations mentioned but translational language remains stronger than evidence 2: Findings are framed as candidates or hypotheses unless clinical utility has been directly tested Reviewer question: Is clinical language matched to the evidence?

Prediction-model rigor 0: Discrimination or Kaplan-Meier separation only 1: Some calibration, DCA, or clinical comparator is reported 2: Locked formula, calibration, DCA, clinical comparator, and intended-use context are all reported Reviewer question: Does the model improve on clinical predictors in an interpretable context?

High-resolution omics rigor 0: Single-cell or spatial data used as decorative confirmation 1: Cell annotation or spatial context discussed with limited validation 2: Integration choices, annotation uncertainty, spatial platform limits, and orthogonal support are explicitly addressed Reviewer question: Are cell states and spatial niches independently supported?

Reproducibility 0: No code, coefficients, cutoffs, package versions, or random seed 1: Partial code or parameters available 2: Executable code, model object, feature list, coefficients, cutoffs, versions, seeds, and derived materials are available Reviewer question: Could the central figure or risk score be reproduced?